CS 422/622 – Project

For undergraduate students (CS422):

- Two make a team.
- A project topic should be interesting.
- Focus on solving interesting problems with appropriate machine learning techniques.
- Performance comparison with other benchmark methods is NOT necessary, but strongly encouraged.

For graduate students (CS622):

- For graduate students, an individual project.
- Performance comparison with other multiple benchmark methods must be presented (e.g. k-fold cross-validation)
- Experimental settings and results must be well presented

The goal of this term project is to apply machine learning techniques appropriately for real-world problems. You can use any libraries or built-in functions.

The project will be evaluated by:

1. Whether the motivation of the project is well described.
2. Whether computational challenges are well described.
3. Whether the dataset is well described.
4. Whether choice of machine learning techniques is well justified.
5. Whether an appropriate approach is applied to realistic data sets that may be being used in practice.
6. How well the experimental settings are described **(especially for graduate students)**
    a. What machine learning methods did you consider for benchmark? How do you tune the optimal parameter?
    b. What are evaluation strategies (e.g., k-fold cross validation) for the performance comparison?
    c. How many times are the experiments repeated?
7. How well the experimental results are discussed.
    a. How can the experimental results be interpreted?
8. Conclusion
9. Misc.
    a. What obstacles have you overcome?

b. Any issues that you want to share with others?
c. Any future plans in related research?

Find a dataset that you may be interested in. Kaggle.com and UCI Machine Learning Repository contain a lot of interesting datasets and well-defined problems. If you have any specific related dataset, you can suggest it too. If you generate a dataset by yourselves, it will be plus (especially for undergraduate students)

Please fill out the form by April 17:

https://docs.google.com/spreadsheets/d/1YT3LFl7fyKSOSzTCTehVxX99JcglMIfEe4iEshXBWdg/edit?usp=sharing

Key dates:

	Date
Forming a team and project title	April 17th
Poster presentation	May 5th
Project submission deadline	May 7th

Rubric

The project final report/poster presentation will be evaluated by:

10. Whether the **motivation** of the project is well described.
11. Whether **computational challenges** are well described. (**CS622**)
12. **Whether the dataset is well described.**
    a. How many samples?
    b. How many features? What types of features?
13. **Is each member's contribution clearly described?**
14. **Whether choice of machine learning techniques is well justified.**
15. Whether an **appropriate approach** is applied to realistic data sets that may be being used in practice.
16. How well the experimental settings are described **(CS622)**
    a. What machine learning methods did you consider for benchmark? How do you tune the optimal parameter?
    b. What are evaluation strategies (e.g., k-fold cross validation) for the performance comparison?
    c. How many times are the experiments repeated?
17. How well the experimental results are discussed.

a. How can the experimental results be interpreted?

18. Conclusion

Notes

+ Presentations will demonstrate your project briefly, but emphasizing project's problem, dataset, challenges, your approaches, and experimental results.

+ The final report submission requires (1) word/pdf file reports (detailed project description and results), (2) source codes.

+ Do not upload data files. Please add the reference (e.g., URL) of the dataset in the final report.